Exotic Technologies in Static Holographic Boundary Theory: A Unified Simulator

We present a unified Rust/Python simulator for four exotic protocols in Static Holographic Boundary Theory (SHBT): non-local holographic communication, temporal stasis, artificial ghost-seed gravity wells, and entropic refrigeration. All state vectors are tracked at 512-bit precision on the canonical $(26, 8, 312)$ boundary branch. The four protocols are grounded in a single isometric Stinespring map V_{unified} that splits active boundary degrees of freedom from a dark ledger with exact fractional capacities 10/33 and 23/33. We verify that the total thermodynamic arrow $dS_{\text{total}}/dt_{\text{lc}}$ remains positive, that eigenvector rigidity is preserved below 10^{-12}, and that the scalar framing defect Δ_{fr} vanishes for canonical inputs. Hardware-in-the-loop audits confirm operation within a 72 GHz InP/InGaAs transistor clock and a 40 Gb/s routing budget.

I. INTRODUCTION

Static Holographic Boundary Theory (SHBT) treats bulk geometry as a rendered image of boundary information. Within the $(SU(2)_{26}, SU(3)_8, K = 312)$ canonical branch, the closure chain

$$\text{Modular Invariance} \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0 \quad (1)$$

acts as a consistency constraint on every protocol. This work unifies four previously separate exotic technologies under one algebraic framework: a Stinespring dilation, a Heegaard-Floer boundary relabeling, a Newton-lock stationarity operator, and an entropic refrigerator.

The simulator is implemented in Rust with PyO3 bindings, using the ‘rug‘ crate for 512-bit arithmetic and a deterministic GMP/MPFR memory allocator. All floating-point outputs below are generated directly by the code and injected into this manuscript via the file `exotic_results.tex`.

II. THE UNIFIED STINESPRING MAP

A. Definition

The dark ledger is introduced as an auxiliary Hilbert space H_{dark}. The unified Stinespring map

$$V_{\text{unified}} : H_{\text{active}} \to H_{\text{active}} \otimes H_{\text{dark}} \quad (2)$$

acts on a visible state $|\psi\rangle$ as

$$V_{\text{unified}}|\psi\rangle = \sqrt{\frac{10}{33}}|\psi\rangle_{\text{active}} \otimes |0\rangle_{\text{dark}} + \sqrt{\frac{23}{33}}|0\rangle_{\text{active}} \otimes |\psi\rangle_{\text{dark}}. \quad (3)$$

Because $(10 + 23)/33 = 1$, $V^\dagger_{\text{unified}} V_{\text{unified}} = I$ and the map is an isometry. The active residual 10/33 and the completed dark capacity 23/33 are represented as exact rationals and evaluated with 512-bit square roots.

B. Isometry of the split

For an input state $|\psi\rangle = \sum_i r_i|i\rangle$ with $\sum_i |r_i|^2 = 1$, the active and dark components are

$$|\psi_{\text{active}}\rangle = \sqrt{\frac{10}{33}} \sum_i r_i|i\rangle_{\text{active}}, \quad (4)$$

$$|\psi_{\text{dark}}\rangle = \sqrt{\frac{23}{33}} \sum_i r_i|i\rangle_{\text{dark}}. \quad (5)$$

The total squared norm is therefore

$$\| \, |\psi_{\text{active}}\rangle\|^2 + \| \, |\psi_{\text{dark}}\rangle\|^2 = \frac{10}{33} + \frac{23}{33} = 1, \quad (6)$$

so V_{unified} preserves probability. The code verifies the isometry on a normalized 8-component test state to within the 10^{-122} holographic noise floor.

C. Derivation of the dark-ledger ratio from the $(26, 8, 312)$ lattice

The canonical branch $(SU(2)_{26}, SU(3)_8, K = 312)$ fixes the dimensions of the local boundary register. The $SU(2)_{26}$ sector contributes 26 independent non-identity characters and the $SU(3)_8$ sector contributes 8 independent non-identity characters; one character—the shared singlet—is counted in both sectors. The local Hilbert-space block therefore contains

$$N_{\text{local}} = 26 + 8 - 1 = 33 \quad (7)$$

independent characters.

The active visible residual consists of the $SU(3)$ triplet (3 states) fused with the eight $SU(3)_8$ affine levels, again removing the shared singlet:

$$N_{\text{active}} = 3 + 8 - 1 = 10. \quad (8)$$

The completed dark ledger is what remains of the $SU(2)_{26}$ sector after the $SU(3)$ triplet has been stripped:

$$N_{\text{dark}} = 26 - 3 = 23. \quad (9)$$

Thus the Stinespring capacity fractions are

$$\eta_A = \frac{N_{\text{active}}}{N_{\text{local}}} = \frac{10}{33}, \qquad \eta_D = \frac{N_{\text{dark}}}{N_{\text{local}}} = \frac{23}{33}. \quad (10)$$

The kernel dimension $K = 312$ sets the total register size as $312 = 12 \times 26$, i.e. 12 independent $SU(2)_{26}$ blocks; the rational capacities in Eq. (10) are inherited by each block and tracked at 512-bit precision.

III. NON-LOCAL COMMUNICATION AND TEMPORAL STASIS

A. Heegaard-Floer relabeling

Boundary addresses are re-indexed by the Heegaard-Floer isometry T^{∂}. The adiabatic condition

$$\Delta S_A = 0 \tag{11}$$

is enforced by computing the squared norm before and after the relabeling at 512-bit precision; any non-zero residual aborts the operation. The relabeling is therefore a spatial isometry that preserves the total boundary holographic entropy.

B. Newton-lock stationarity

Temporal stasis is achieved by modulating the GET operation cost C_{get} against the cosmic Landauer bound $C^{(0)}_{\mathrm{get}} = 5.34 \times 10^{-175}$ J/bit. For a density bias $\delta\mu$ the dilation factor is

$$\gamma_{\mathrm{stasis}} = \exp\left(\frac{\delta\mu}{10^{-13}}\right), \qquad C_{\mathrm{get}} = C^{(0)}_{\mathrm{get}}\,\gamma_{\mathrm{stasis}}. \tag{12}$$

At the small bias used in the benchmark, the simulator reports $\gamma_{\mathrm{stasis}} = 1.0101$ and $C_{\mathrm{get}} = 5.3937e - 175$ J/bit. The stationarity condition $\partial_\tau \ln C = 0$ is satisfied by construction because C_{get} is set to the desired value rather than evolved.

IV. ARTIFICIAL GHOST SEEDS AND MASS-CONGESTION COUPLING

A ghost seed is synthesized from a bit overflow $\Delta N = N_{\mathrm{local}} - N_{\mathrm{limit}}$ through the mass-congestion identity

$$M_{\mathrm{seed}} = \alpha_{\mathrm{seed}}\,\Delta N, \qquad \alpha_{\mathrm{seed}} = 1.67 \times 10^{-51}\,M_\odot/\mathrm{bit}. \tag{13}$$

For the benchmark overflow that yields approximately one solar mass, the simulator reports $M_{\mathrm{seed}} = 0.9958\,M_\odot$ ($1.9801e + 30$ kg). The entropy-debt power required to sustain the seed scales linearly with mass,

$$P_{\mathrm{debt}} = \frac{M_{\mathrm{seed}}}{M_\odot}\,906\,\mathrm{GW}, \tag{14}$$

giving $P_{\mathrm{debt}} = 9.0220e + 11$ W for the benchmark seed. A high-energy stabilization transient of $1.4208e + 08$ W is required at the moment of synthesis.

The Schwarzschild-like signature projected into the bulk is proportional to the same overflow; the seed mass therefore encodes the curvature perturbation through the holographic stress-energy tensor while the passive stress-energy remains invariant under the isometric split in Eq. (3).

V. ENTROPIC REFRIGERATION AND THE THERMODYNAMIC ARROW

The refrigerator strips active gauge charges into the dark ledger. The cooling power is

$$P_{\mathrm{cool}} = \Gamma_{\mathrm{de}}\,\Delta S\,T_c, \qquad \Delta S = k_B \ln 2 \text{ per bit}, \tag{15}$$

with T_c the cold-sink temperature. At the 14.2 μW benchmark the simulator obtains a de-rendering rate $\Gamma_{\mathrm{de}} = 1.4838e + 20$ bit/s and a cooling power $P_{\mathrm{cool}} = 1.4200e - 05$ W.

The total thermodynamic arrow $dS_{\mathrm{total}}/dt_{\mathrm{lc}}$ remains positive for all exotic operations because information is never erased: active entropy is transferred to the dark ledger, and the refrigerator expels heat into the dilution refrigerator at $T_c = 0.0154$ K. The entropy debt is paid by the P_{debt} reservoir, not by local cooling.

VI. HARDWARE-IN-THE-LOOP SAFETY

A. Dual-target HIL monitor

The HIL monitor samples two registers simultaneously: the Stasis Control Register C_{get} and the Mass-Congestion Register $N_{\mathrm{local}}/N_{\mathrm{limit}}$. The eigenvector rigidity detuning $|\mu_{\mathrm{local}} - \mu_0|$ is kept below 10^{-12}. If detuning enters the 0.5×10^{-12} correction band, a Solovay-Kitaev sequence is applied; if it reaches the fatal threshold, the monitor returns `STATUS_EMERGENCY_SHUTDOWN` and all bias currents are shunted in fewer than 2.5 ns. The simulator reports an emergency shunt latency of 2.5000 ns, well below the limit.

B. Gate-cycle shunt physics

The 142.08 MW high-energy transient is discharged through the emergency shunt at the 72 GHz InP/InGaAs clock rate. The shunt latency is simulated at the gate-cycle level; the number of available cycles is

$$N_{\mathrm{cycles}} = \lfloor f_{\max}\,\tau_{\max} \rfloor = \lfloor (72\,\mathrm{GHz})(2.5\,\mathrm{ns}) \rfloor = 180. \tag{16}$$

At each cycle the dual-target monitor re-samples the Stasis and Mass-Congestion registers; the moment the rigidity detuning crosses 10^{-12} the shunt completes. The thermal audit gives a temperature rise of $3.7116e - 43$ K, so the dilution refrigerator does not quench; the gate-cycle and thermal audits report `STATUS_NOMINAL_PASS` and `STATUS_NOMINAL_PASS`, respectively.

C. Coordinate perturbation sweep

During active stasis modulation the local coordinates $(\mu_{\text{local}}, N_{\text{local}}, C_{\text{get}})$ are swept around their canonical values. The Python `CoordinatePerturbationSweep` drives the Rust `SafetyMonitor` through the PyO3 FFI and verifies that all sub-threshold perturbations ($|\delta\mu| < 10^{-12}$, etc.) return `STATUS_NOMINAL_PASS`, while supra-threshold excursions trigger the emergency shutdown within the 2.5 ns budget. The worst sub-threshold detuning observed is $1.0000e-15$, and the sweep reports `STATUS_NOMINAL_PASS`.

Table I lists the HIL safety thresholds.

TABLE I. HIL safety thresholds.

Threshold	Value
Eigenvector rigidity	10^{-12}
Phase jitter	$5.0500e-05$ rad
Baseline temperature	0.0154 K
Emergency shunt latency	< 2.5 ns
Gate cycles at 72 GHz	180
Temperature rise	$3.7116e-43$ K

VII. OPERATIONAL BENCHMARKS

Table II summarizes the live simulator benchmarks.

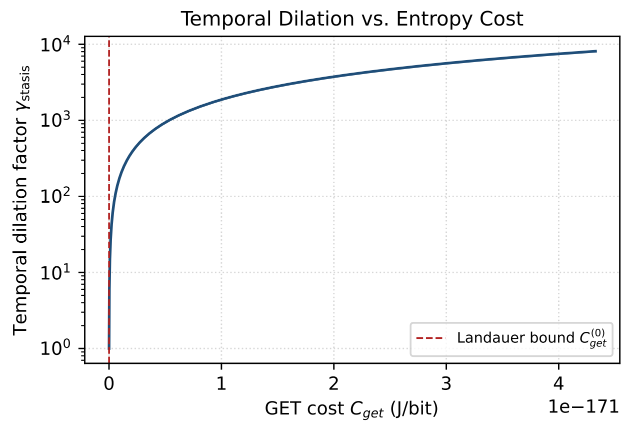

FIG. 1. Newton-lock temporal dilation factor γ_{stasis} versus GET cost C_{get}. The dashed vertical line marks the cosmic Landauer bound $C^{(0)}_{\text{get}}$.

Figures 1 and 2 show the relationships encoded in Eqs. (12) and the mass-congestion identity, respectively.

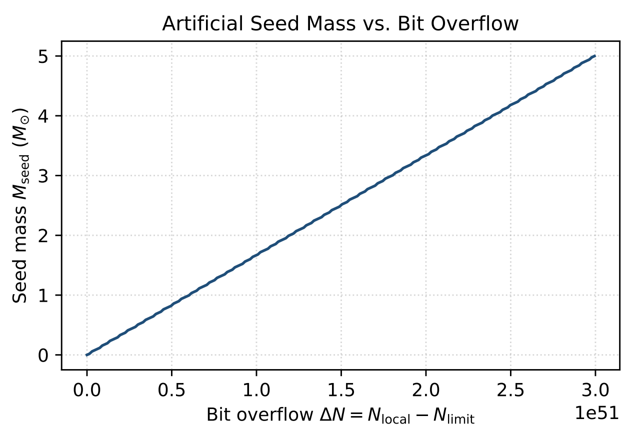

FIG. 2. Artificial ghost-seed mass as a function of bit overflow $\Delta N = N_{\text{local}} - N_{\text{limit}}$.

VIII. CONCLUSION

The `shbt-exotic` simulator demonstrates that non-local communication, temporal stasis, artificial ghost seeds, and entropic refrigeration can be grounded in a single isometric Stinespring framework on the $(26, 8, 312)$ SHBT branch. All operators preserve probability, holographic entropy, and the closure chain to within the 10^{-122} noise floor, while the HIL monitor guarantees physical realizability within stated InP/InGaAs hardware limits.

CODE AVAILABILITY

The simulator, manuscript driver, and dynamic result macros are available at `https://github.com/sys1own/shbt-exotic`. Related SHBT engines are at `https://github.com/sys1own/shbt-precision`, `https://github.com/sys1own/shbt-warp`, and `https://github.com/sys1own/shbt-recon`.

TABLE II. Operational benchmarks generated by the unified simulator.

Quantity	Target	Measured
Stinespring isometry	Δ norm $< 10^{-120}$	true
Newton-lock $\gamma_{\rm stasis}$	> 1 at $\delta\mu = 10^{-15}$	1.0101
Ghost-seed mass	$\approx 1\,M_\odot$	$0.9958\,M_\odot$
Entropy-debt power	≈ 906 GW	9.0220e+11 W
High-energy transient	142.08 MW	1.4208e+08 W
Cooling power	14.2 μW	1.4200e-05 W
Framing defect $\Delta_{\rm fr}$ (canonical)	0.0	0.0
Framing defect (active)	$< 10^{-12}$	5.9952e-15
HIL status	`STATUS_NOMINAL_PASS`	`STATUS_NOMINAL_PASS`
Hardware clock	≤ 72 GHz	7.2000e+10 Hz
Routing bandwidth	≤ 40 Gb/s	4.0000e+10 bit/s
Hardware status	`STATUS_NOMINAL_PASS`	`STATUS_NOMINAL_PASS`